PROJ. REFERENCE:	DRAWING NAME:		REVISION:		DRAWN BY:	
PJR-2026-003	GENERAL NOTE PART-01 (GENERAL / DESIGN BASIS & LOADING)		R00		P.LIHENG	N
DRAWING NUMBER:	DRAWING STAGE:	CLIENT:	SCALE:		ISSUE DATE:	
WH-STR-SD-003	FOR APPROVAL	Mr. XXX	N.T.S.		13-OCT-2026	STRUCTURAL (BIM)

A. GENERAL

1. THESE NOTES SHALL BE READ WITH ALL ARCHITECTURAL, MEP, EQUIPMENT (STEAM BOILER) AND CLADDING SUPPLIER DRAWINGS. ANY DISCREPANCY SHALL BE REPORTED TO THE STRUCTURAL ENGINEER BEFORE WORK PROCEEDS.
2. ALL DIMENSIONS ARE IN mm AND LEVELS IN m UNLESS NOTED. DO NOT SCALE FROM DRAWINGS. WRITTEN DIMENSIONS GOVERN.
3. LEVEL +0.000 = FINISHED GROUND FLOOR LEVEL (FFL) OF THE WAREHOUSE. STEEL BASE PLATES SIT ON THE STUMP COLUMNS AT +0.000 (FFL); TOP OF BRICK WALL = +2.000. EAVE = +12.000. RIDGE = +12.656.
4. THE STRUCTURE IS A SINGLE-SPAN STEEL PORTAL FRAME (15.0 m SPAN, 5 FRAMES AT 6.25 m = 25.0 m) ON RC STUMP COLUMNS (-0.500 TO +0.000), PILE CAPS AND PRECAST DRIVEN RC PILES 300x300x7000 (2 PCS PER CAP). ROOF SLOPE 5 deg.
5. ROOF AND WALL ABOVE +2.000 = 40 mm PU SANDWICH PANEL. PERIMETER WALL = 200 mm BRICK + PLASTER BOTH SIDES, H = 2.0 m FROM FFL, STIFFENED BY C2 COLUMNS (200x200 @ 3125) + BOND BEAM BB1.
6. THE BRICK WALL, GROUND BEAMS AND GROUND SLAB ARE NON-STRUCTURAL WITH RESPECT TO THE STEEL FRAME. THE STEEL FRAME ACTS ALONE FOR ALL ROOF AND WIND LOADS.
7. THE CONTRACTOR SHALL VERIFY SITE LEVELS, SETBACKS AND BOUNDARY CLEARANCES (PCP1 PROJECTS 1.70 m FROM GRID A/B, PCP2 1.20 m FROM GRID 1/5) BEFORE EXCAVATION.
8. TEMPORARY WORKS AND STABILITY OF THE PARTLY ERECTED STEELWORK ARE THE CONTRACTOR'S RESPONSIBILITY. TEMPORARY BRACING SHALL REMAIN UNTIL ROOF BRACING AND CLADDING ARE COMPLETE.

B. DESIGN CODES

CODE	SCOPE
EN 1990	BASIS OF STRUCTURAL DESIGN
EN 1991-1-1	SELF-WEIGHT, IMPOSED LOADS
EN 1991-1-3	SNOW
EN 1991-1-4	WIND
EN 1992-1-1	CONCRETE STRUCTURES (RC)
EN 1993-1-1	STEEL - GENERAL RULES
EN 1993-1-3	COLD-FORMED MEMBERS (PURLINS/GIRTS)
EN 1993-1-5	PLATED STRUCTURAL ELEMENTS
EN 1993-1-8	JOINTS
EN 1997-1	GEOTECHNICAL DESIGN
EN 10025-2, EN 1090-2	STEEL PRODUCTS, EXECUTION (EXC2)
EN 206, EN 13670	CONCRETE, EXECUTION OF CONCRETE
EN 1996-1-1	MASONRY (INFILL WALL, ROBUSTNESS ONLY)

PER SD-003: CEN RECOMMENDED VALUES USED (NO NATIONAL ANNEX FOR CAMBODIA); LOCAL AUTHORITY REQUIREMENTS APPLY WHERE MORE ONEROUS. CONFIRMED BY USER (P.LIHENG) 14-OCT-2026.

E. DESIGN RESULTS SUMMARY (PF1)

ITEM	RESULT	LIMIT
COLUMN TAPERED WELDED I	UTIL. 0.30	1.00
RAFTER IPE600 S355 (CLASS 1)	UTIL. 0.79	1.00
MEd EAVE / RIDGE (ULS)	835 / 292 kNm	-
EAVE SWAY (WIND, SLS)	114 mm	h/100 = 120 mm
RAFTER DEFLECTION (Q)	2 mm	L/200 = 37 mm
PILE CAP PCP1 2400x1000x500 (x10)	2 PILES 300x300x7000	OK
PILE CAP PCP2 2400x1000x500 (x2)	2 PILES 300x300x7000	OK
PILE COMPRESSION (G+Q / G+W)	69 / 101 kN	Qa 93 / 116 kN
PILE UPLIFT (FRAME / BRACED BAY)	61 / 63 kN	Qt 65 kN (ASSUMED)
PILE LATERAL (SERVICE)	53 kN	68 kN (ASSUMED)

C. DESIGN BASIS & ACTIONS

ACTION	VALUE	SOURCE
DESIGN WORKING LIFE	50 YEARS (CAT. 4)	EN 1990 T2.1
CONSEQUENCE CLASS	CC2 (KFI = 1.0)	EN 1990 B
ROOF DEAD G (PANEL+PURLIN+BRACE+MEP)	0.50 kN/m2	EN 1991-1-1
STEEL FRAME SELF-WEIGHT	COMPUTED x 1.15	CONNECTIONS
WALL CLADDING + GIRTS	0.17 kN/m2	SUPPLIER TBC
ROOF IMPOSED Q (CAT. H, SLOPE < 15 deg)	0.40 kN/m2 (PLAN)	EN 1991-1-1 6.3.4
BASIC WIND SPEED vb	33.0 m/s	GIVEN
TERRAIN CATEGORY	II (z0 = 0.05 m)	ASSUMED
PEAK VELOCITY PRESSURE qp(12.66)	1.71 kN/m2	EN 1991-1-4 4.8
cpe WINDWARD / LEEWARD WALL (D / E)	+0.78 / -0.46	T7.1 (h/d=0.85)
cpe ROOF G / H / I / J	-1.17/-.58/-.59/-.63	T7.4a
INTERNAL PRESSURE cpi	+0.2 / -0.3	7.2.9 (MAX. CASE)
SEISMIC	NEGLIGIBLE (PHNOM PENH)	LOW SEISMICITY
THERMAL / BOILER LOADS	NOT INCLUDED	TO BE ADVISED

D. LOAD COMBINATIONS (EN 1990)

1. ULS (STR/GEO) EQ. 6.10: 1.35 G + 1.5 Q ; 1.35 G + 1.5 W ; 1.0 G + 1.5 W (UPLIFT) ; 1.35 G + 1.5 Q + 0.9 W (psi0 = 0.6, 1.5 x 0.6 = 0.9).
2. EQUIVALENT HORIZONTAL FORCES (SWAY IMPERFECTION phi = 1/346, PHI0 = 1/200, alpha_h = 0.58, alpha_m = 0.83 -> 0.0029) ADDED TO ALL ULS COMBINATIONS. 2nd-ORDER (P-DELTA) ANALYSIS PERFORMED.
3. SLS CHARACTERISTIC: G + Q ; G + W. EAVE SWAY LIMIT h/100 = 120 mm UNDER WIND (PROJECT CRITERION, PU PANEL CLADDING - TBC). RAFTER VERTICAL DEFLECTION LIMIT L/200 (Q ONLY).
4. PILES: WORKING-STRESS CHECK ON SERVICE LOADS, EN 1997-1 6.5.2 / FoS = 2.5. PILE CAP SELF-WEIGHT + STUMP INCLUDED (35 kN); LEVER ARM 1.0 m; PILE COUPLE s = 1.8 m.
5. PILE DATA (GIVEN): PRECAST SQUARE DRIVEN PILE 300x300, L = 7.0 m, Qu = 232 kN, FoS = 2.5 -> Qa = 93 kN. TENSION Qt = 65 kN AND LATERAL H = 68 kN PER PILE ARE ASSUMED - TBC BY PILE LOAD TESTS.

F. ASSUMPTIONS TO BE CONFIRMED

F1. FOUNDATION - PRECAST SQUARE DRIVEN PILES 300X300, L=7.0 M; SOIL INVESTIGATION AND PILE LOAD TESTS REQUIRED
F2. BOILER / PIPING / CRANE / HOIST AND ANY PLATFORM LOADS NOT INCLUDED
F3. OPENINGS (DOORS, LOUVRES, FLUE) NOT SHOWN
F4. TERRAIN CATEGORY II AND NO SHIELDING, SITE ALTITUDE < 100 M
F5. LONGITUDINAL (GRID A/B) WIND SWAY BY BRACED BAYS - CONFIRM
F6. PURLINS/GIRTS COLD-FORMED - SUPPLIER VERIFICATION FOR LOADS IN SD-108
F7. CLIENT NAME IS A PLACEHOLDER
F8. EAVE SWAY LIMIT H/100 = 120 MM - CONFIRM WITH CLADDING SUPPLIER

PROJ. REFERENCE:	DRAWING NAME:		REVISION:		DRAWN BY:	
PJR-2026-003	GENERAL NOTE PART-02 (CONCRETE & REINFORCEMENT)		R00		P.LIHENG	N
DRAWING NUMBER:	DRAWING STAGE:	CLIENT:	SCALE:		ISSUE DATE:	
WH-STR-SD-004	FOR APPROVAL	Mr. XXX	N.T.S.		13-OCT-2026	STRUCTURAL (BIM)

A. MATERIALS (CONCRETE & REINFORCEMENT)

ELEMENT	CLASS (EN 206)	EXPOSURE	NOTES
LEAN CONCRETE (50 mm)	C12/15	X0	BLINDING
PILE CAPS, TB1, GROUND BEAM	C30/37	XC2	Dmax 20, S3
PRECAST PILES 300x300x7000	C40/50 MIN.	XC2	FACTORY, EN 12794
STUMP PC1/PC3, COLUMN C2	C30/37	XC3 / XC4	Dmax 20, S3
GROUND SLAB 150	C25/30 MIN. (C30/37)	XC3	S2, w/c <= 0.55
REINFORCEMENT	fyk = 390 MPa (GRADE 390)	-	EN 10080 DUCT. B
ANCHOR BOLTS	DB24 GR.390 THREADED, HDG	-	8 PER BASE
GROUT (BASE PLATE)	NON-SHRINK, fck >= 50 MPa	-	40 mm THICK

CEMENT: PORTLAND CEMENT CEM I / II (EN 197-1). NO CALCIUM CHLORIDE. MAX. WATER-CEMENT RATIO 0.55.

B. CONCRETE COVER (EN 1992-1-1 4.4)

ELEMENT	Cnom (mm)
FOUNDATION / PILE CAP (CAST AGAINST LEAN)	50
FOUNDATION / PILE CAP TOP & SIDES (BACKFILLED)	50
PRECAST PILE (TO LINKS)	40
COLUMNS / STUMPS (TO LINKS)	40
GROUND BEAMS / BOND BEAMS (TO LINKS)	40
SLABS (TOP / BOTTOM MESH)	40 (MIN. 30)
SPACER BLOCKS (CONCRETE) AT 1.0 m C/C MAX.	REQUIRED

Cnom = Cmin + Delta_c,dev (10 mm). COVER TO BE CONFIRMED AGAINST SITE EXPOSURE (MARINE / CHEMICAL EXPOSURE RAISES THE CLASS).

C. ANCHORAGE & LAP LENGTHS (C30/37, fyk = 390)

BAR phi (mm)	10	12	16	20	25
lbd GOOD BOND (mm)	280	335	445	555	695
lbd POOR BOND (mm)	400	475	635	795	995
LAP l0 GOOD, <=50% LAPPED	420	500	670	830	1040
LAP l0 POOR, <=50% LAPPED	600	710	950	1190	1490
HOOK / BEND MIN. MANDREL (mm)	40	48	64	140	175

lb,rqd = (phi/4)(sigma_sd / fbd) ; fbd = 2.25 eta1 eta2 fctd = 3.04 MPa (GOOD BOND). eta1 = 0.7 FOR POOR BOND (TOP BARS IN MEMBERS > 250 mm DEEP). alpha_6 = 1.5 (50% LAPPED). MANDRELS: 4 phi (phi <= 16), 7 phi (phi > 16).

D. REINFORCEMENT NOTES

1. BAR TYPE: DB = DEFORMED BAR (fyk = 390 MPa); RB / R = PLAIN BAR (LINKS OF 6 mm ONLY IF DEFORMED BAR NOT AVAILABLE). NO BAR SHALL BE REBENT ON SITE EXCEPT AS PERMITTED BY EN 13670.
2. FOUNDATION / PILE CAP TOP BARS PROVIDED WHERE UPLIFT OCCURS. PILE MAIN BARS ANCHORED INTO CAP PER DETAIL DRAWINGS.
3. LAPS: STAGGER SO THAT NOT MORE THAN 50% OF BARS ARE LAPPED AT ONE SECTION. NO LAPS AT LOCATIONS OF MAX. MOMENT. BARS > DB32 SHALL NOT BE LAPPED.
4. LINKS: CLOSED, 135 deg HOOKS WITH 10 phi (MIN. 75 mm) STRAIGHT EXTENSION, ALTERNATING SEATING.
5. MINIMUM SPACING BETWEEN BARS = MAX(phi, 20 mm, Dmax + 5 mm).
6. STARTER BARS & ANCHOR BOLTS SHALL BE FIXED BY TEMPLATE AND SECURED AGAINST DISPLACEMENT; TOLERANCE +/- 3 mm.
7. FOR GROUND BEAMS AND BOND BEAMS AT WALL CORNERS PROVIDE L-BARS (LAP 50 phi).

E. CONCRETING & CURING

1. CONCRETE SHALL BE COMPACTED WITH POKER VIBRATORS. NO ADDITION OF WATER ON SITE. MIN. 7 DAYS WET CURING (EN 13670 CURING CLASS 2).
2. COMPRESSION TESTING: 1 SET OF 3 SAMPLES PER 50 m3 OR PER POUR (7 / 28 DAYS) TO EN 12390.
3. CONSTRUCTION JOINTS: REMOVE LAITANCE AND ROUGHEN TO 5 mm AMPLITUDE; WET BEFORE NEXT POUR. PROVIDE 50 mm KEYWAY WHERE REQUIRED.

F. EARTHWORKS & GROUND SLAB

1. REMOVE TOP SOIL AND ORGANIC MATERIAL. PROOF-ROLL SUB-GRADE AND COMPACT FILL TO >= 95% MODIFIED PROCTOR (MDD) IN 200 mm MAX. LAYERS.
2. FORMATION LEVELS SHALL BE INSPECTED BY THE ENGINEER. REMOVE SOFT SPOTS AND REPLACE WITH LEAN CONCRETE.
3. GROUND SLAB BUILD-UP: RC SLAB ON 0.2 mm POLYTHENE SLIP MEMBRANE (LAPPED 300) - 50 mm LEAN CONCRETE - 100 mm CRUSHED STONE - COMPACTED FILL.
4. CONTRACTION (SAWCUT) JOINTS AT 4.0 m x 4.0 m MAX. (<= 25 x SLAB DEPTH), DEPTH 1/4 SLAB, CUT WITHIN 12-24 h AND SEALED.
5. ISOLATION JOINTS (20 mm COMPRESSIBLE FILLER + SEALANT) AT ALL COLUMNS, WALLS AND EQUIPMENT PLINTHS.

G. BRICKWORK

1. BRICK WALL 200 mm, CEMENT MORTAR 1:4, PLASTERED BOTH SIDES. PROVIDE RC STIFFENER COLUMNS C2 (200x200 @ ~3.0 m MAX.) AND BOND BEAMS BB1 (200x200).
2. MASONRY WALLS ARE NON-LOAD-BEARING. ISOLATE TOP OF WALL FROM PRIMARY STRUCTURE BY 20 mm GAP + FLEXIBLE SEALANT. TIE WALL TO COLUMNS WITH DB8 DOWELS @ 600 (L=500).

PROJ. REFERENCE:	DRAWING NAME:		REVISION:		DRAWN BY:	
PJR-2026-003	GENERAL NOTE PART-03 (STRUCTURAL STEELWORK)		R00		P.LIHENG	N
DRAWING NUMBER:	DRAWING STAGE:	CLIENT:	SCALE:		ISSUE DATE:	
WH-STR-SD-005	FOR APPROVAL	Mr. XXX	N.T.S.		13-OCT-2026	STRUCTURAL (BIM)

A. MATERIALS

ITEM	SPECIFICATION
PLATES, WELDED & HOT-ROLLED	S355JR (EN 10025-2) ; fy = 355 MPa (t <= 40)
SECONDARY CHS BRACING STRUTS	S355J0H (EN 10210) / EN 10219
PURLINS & GIRTS (Z200x3.0)	S350GD+Z275 (EN 10346), SUPPLIER DESIGN
EAVE GUTTER / DOWNPIPES	GALV. t=1.2 (Z275), 350x200, FALL 1:200; D100
BOLTS (STRUCTURAL)	CLASS 8.8 / 10.9 (EN 14399-3 HR/HV PRELOADED)
ANCHOR BOLTS	DB24 GR.390 THREADED, GALV., 8/BASE, EMBED 450
BRACING RODS	WALL D30 / ROOF D20, CLASS 8.8 + TURNBUCKLES
WELDING CONSUMABLES	E70xx / ER70S (EN ISO 14341), MATCHING S355

B. SECTIONS ADOPTED

MARK	SECTION	USE	CLASS
PF1 COL	WELDED I TAPER 450->1100 x 300 FL25/W10	PORTAL COLUMN	3
PF1 RAF	IPE600 (HAUNCH CUT FROM IPE600)	PORTAL RAFTER	1
WP1	IPE500	GABLE WIND POST (1/ENDWALL)	1
P1	Z200 x 3.0 @ 1507 (SLOPE)	PURLIN	-
G1	Z200 x 3.0 @ 1500	SIDE-WALL GIRT	-
G2	Z200 x 3.0 @ 1000 (L = 7.5 m)	GABLE-WALL GIRT	-
ES1	CHS 114.3x4	EAVE STRUT / STRUT	-
WB	D30 ROD	WALL X-BRACE (BAYS 1-2, 4-5)	-
RB	D20 ROD (M20 8.8)	ROOF X-BRACE (BAYS 1-2, 4-5)	-

E. ERECTION

1. ERECTION SEQUENCE: BASE PLATES ON SHIMS (40 mm GROUT SPACE) -> COLUMN LIFT -> RAFTER -> ROOF / WALL BRACING IN END BAYS -> EAVE STRUTS -> PURLINS & GIRTS -> CLADDING. FRAMES SHALL NOT BE LEFT UNBRACED.
2. TENSION RODS: SNUG-TIGHT TURNBUCKLE ONLY; NO PRESTRESS. CHECK LINE AND LEVEL BEFORE FINAL BOLT TIGHTENING.

C. FABRICATION & WELDING

1. FABRICATION AND ERECTION TO EN 1090-2 EXECUTION CLASS EXC2. FACTORY PRODUCTION CONTROL TO EN 1090-1.
2. WELDS: CONTINUOUS FILLET 6 mm MIN. UNLESS NOTED; FULL-PENETRATION BUTT (CJP) FOR FLANGE-TO-END-PLATE AND ALL SPLICES. WELDERS QUALIFIED TO EN ISO 9606-1; WPS TO EN ISO 15614-1.
3. NDT: 100% VISUAL; 10% MPI ON FILLET WELDS; 100% UT OF CJP BUTT WELDS AT FLANGE SPLICES, MIN. 20% OTHER (EXC2: ISO 17637 / 17640).
4. BUILT-UP WELDED I-SECTIONS: DOUBLE-SIDED 6 mm FILLET (a = 4.2 mm) MIN. WEB-TO-FLANGE WELD, CHECKED FOR SHEAR FLOW.
5. HOLES FOR M24 = 26 mm, M20 = 22 mm, M16 = 18 mm. EDGE DISTANCE >= 1.2 d0 (>= 40 mm FOR M24), PITCH >= 2.2 d0.
6. PRELOADED CONNECTIONS: HR / HV 10.9 BOLTS PRELOADED TO 0.7 fub As (M24: 220 kN) BY TORQUE-CONTROL OR PART-TURN. FAYING SURFACES FLAT AND UNPAINTED.
7. FLY BRACES AT MAX. 3.0 m C/C ALONG COMPRESSION FLANGES OF RAFTERS AND COLUMNS.
8. FABRICATION TOLERANCES EN 1090-2 CLASS 1; ERECTION TOLERANCES: COLUMN PLUMB h/300 (MAX. 20 mm), FRAME SPACING +/- 5 mm.
9. NO FIELD WELDING OF PRIMARY MEMBERS WITHOUT ENGINEER'S APPROVAL. NO MEMBER CUT OR DRILLED OUTSIDE APPROVED SHOP DRAWINGS.
10. CONTRACTOR TO SUBMIT SHOP DRAWINGS AND CONNECTION CALCULATIONS (EN 1993-1-8) FOR APPROVAL BEFORE FABRICATION.

D. CORROSION PROTECTION

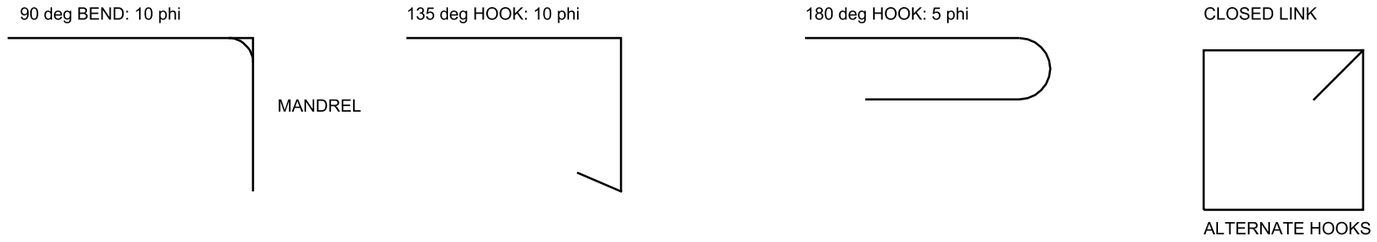
1. SURFACE PREPARATION Sa 2.5 (ISO 8501-1). SYSTEM C3 (MEDIUM, 15 YEARS, ISO 12944): 80 um EPOXY ZINC-RICH PRIMER + 120 um POLYURETHANE TOPCOAT (TOTAL 200 um). BOLTS HOT-DIP GALVANISED.
2. NO PAINT ON AREAS TO BE WELDED OR CONTACT SURFACES OF PRELOADED BOLTS (MASKED).
3. PURLINS / GIRTS: Z275 GALVANISED. BASE PLATE AND ANCHOR BOLTS: HOT-DIP GALVANISED (EN ISO 1461).
4. CORROSIVITY CATEGORY TO BE CONFIRMED BY CLIENT AGAINST INTERNAL HUMIDITY / CHEMICAL ENVIRONMENT.

F. CONNECTION DESIGN BASIS

PF1 EAVE (M = 835 kNm, V = 174 kN, N = -31 kN), RIDGE (M = 292 kNm, WITH BOTTOM HAUNCH PLATE): BOLTED END PLATE WITH 10.9 BOLTS, T-STUB ANALYSIS EN 1993-1-8 6.2.4, COLUMN WEB STIFFENERS AS DETAILED ON SD-107. DETAILS GIVEN ARE INDICATIVE AND SUBJECT TO THE FABRICATOR'S VERIFICATION.

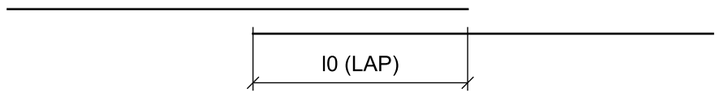
PROJ. REFERENCE:	DRAWING NAME:		REVISION:		DRAWN BY:	
PJR-2026-003	GENERAL NOTE PART-04 (RC STANDARD DETAILS)		R00		P.LIHENG	N
DRAWING NUMBER:	DRAWING STAGE:	CLIENT:	SCALE:		ISSUE DATE:	
WH-STR-SD-006	FOR APPROVAL	Mr. XXX	N.T.S.		13-OCT-2026	STRUCTURAL (BIM)

1 STANDARD BENDS AND HOOKS (EN 1992-1-1 8.3, EN 13670)



BAR phi (mm)	MIN. MANDREL (HOOKS/BENDS)	LINKS (phi <= 16)	BEND FOR LAPS (phi > 16)
8 - 16	4 phi	4 phi	-
20	7 phi	-	7 phi
25	7 phi	-	7 phi
32	7 phi	-	7 phi

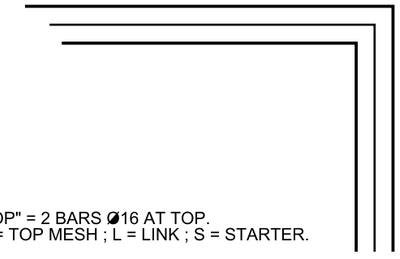
2 LAP SPLICE (TENSION) AND ANCHORAGE



l0 = alpha6 lbd >= l0,min = max(0.3 alpha6 lb,rqd ; 15 phi ; 200 mm). TRANSVERSE SPACING <= 4 phi OR 50 mm.

LAP LENGTHS: SEE SD-004 TABLE C. NO LAPS WITHIN 2 x h OF BEAM / COLUMN INTERFACES.

3 TYPICAL CORNER DETAIL - GB1 / BB1 (PLAN)



4 REINFORCEMENT NOTATION

"12 DB20" = 12 BARS ⌀20 mm DEFORMED BAR ; "DB10-150" = ⌀10 AT 150 mm C/C ; "2DB16 TOP" = 2 BARS ⌀16 AT TOP.
BARS ARE MEASURED TO OUTSIDE OF BEND. BAR MARKS: B1, B2 = BOTTOM MESH ; T1, T2 = TOP MESH ; L = LINK ; S = STARTER.

RC STANDARD NOTES

1. ALL DETAILS FOLLOW EN 1992-1-1 AND EN 13670; IN CASE OF DOUBT CONSULT THE ENGINEER.
2. LINKS SHALL BE TIED WITH 1.6 mm ANNEALED WIRE AT EVERY BAR INTERSECTION IN COLUMNS / STUMPS (ALTERNATE ELSEWHERE).
3. BARS SHALL BE FREE FROM LOOSE RUST, OIL AND MUD AT TIME OF CONCRETING. NO RE-BENDING OF PARTLY EMBEDDED BARS WITHOUT APPROVAL.
4. MINIMUM CLEAR SPACING = MAX(phi, 20 mm, Dmax + 5 mm = 25 mm). BARS IN COLUMNS: MIN. 40 mm.
5. SPACER BLOCKS: CONCRETE, SAME GRADE AS ELEMENT (PILE CAPS: 50 mm, 4 No./m2). CHAIRS: STEEL, GALV. FOR SLAB MESH AT 1000 c/c EACH WAY.
6. STARTER BARS ARE FIXED BY TEMPLATE AND SECURED AGAINST DISPLACEMENT DURING POURING. STARTER LAP = lbd x 1.5 (SD-004).
7. CONSTRUCTION JOINTS: ROUGHEN SURFACE TO 5 mm AMPLITUDE, REMOVE LAITANCE; WET BEFORE NEXT POUR. KEYWAY 50 mm.
8. EXPOSED STUMP EDGES: 20 mm CHAMFER. TOP OF STUMP LEVEL (+0.000) UNDER THE 40 mm BASE GROUT; GROUT SLOPED 1:20 AWAY FROM THE BASE PLATE.
9. STRIKE-OFF OF FORMWORK: SIDES 24 h MIN. (>= 5 MPa), SOFFITS 7 DAYS.
10. EMBEDMENTS (ANCHOR BOLTS, SLEEVES) SHALL BE FIXED BEFORE CONCRETING AND CHECKED BY SURVEY: TOLERANCE +/- 3 mm ON ANCHOR BOLTS, +/- 10 mm ON STUMP SETTING-OUT.

PROJ. REFERENCE:	DRAWING NAME:		REVISION:		DRAWN BY:	
PJR-2026-003	GENERAL NOTE PART-05 (STEEL STANDARD DETAILS)		R00		P.LIHENG	N
DRAWING NUMBER:	DRAWING STAGE:	CLIENT:	SCALE:		ISSUE DATE:	
WH-STR-SD-007	FOR APPROVAL	Mr. XXX	N.T.S.		13-OCT-2026	STRUCTURAL (BIM)

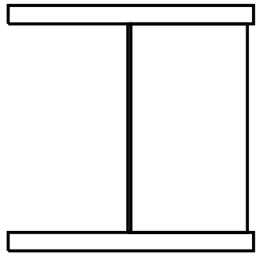
1 BOLT HOLE & SPACING TABLE (EN 1993-1-8 T3.3)

BOLT	d0 (HOLE)	e1 (EDGE)	p1 (PITCH)	TIGHTENING	NOTE
M12	14	>= 1.2 d0 = 17	>= 2.2 d0 = 31	SNUG	SECONDARY
M16	18	>= 22	>= 40	SNUG	GIRT CLEATS
M20	22	>= 27	>= 49	SNUG (8.8)	BRACING, FIN PL
M24	26	>= 31 (40 SPEC.)	>= 57 (70 SPEC.)	PRELOAD 10.9 (220 kN)	PRIMARY JOINTS

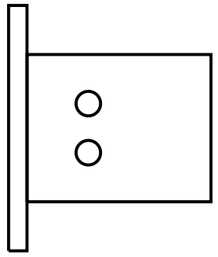
2 WELD SYMBOLS / SIZES

SYMBOL	DESCRIPTION	TYPICAL USE
6 [FILLET]	CONTINUOUS FILLET 6 mm (BOTH SIDES UNLESS NOTED)	WEB-FLANGE (a = 4.2), STIFFENERS
8 [FILLET]	CONTINUOUS FILLET 8 mm	BASE PL / COLUMN
CJP	COMPLETE-JOINT-PENETRATION BUTT	FLANGE SPLICES, END PLATE TO FLANGE
[ALL-ROUND]	ALL-ROUND WELD	GUSSET, FIN PL
FW	FIELD WELD (ONLY WITH APPROVAL)	-

3 TYPICAL STIFFENER / FIN PLATE (SCHEMATIC)



STIFFENER PL12 (BOTH SIDES),
CLIPPED 20 x 20 AT CORNERS;
FILLET 6 mm TO WEB AND FLANGE



FIN PL10, 2 x M20 (8.8),
WELD 6 mm ALL ROUND TO FLANGE

4 ABBREVIATIONS

PL = PLATE ; FL = FLANGE ; CJP = COMPLETE JOINT PENETRATION ; HR/HV = HIGH-STRENGTH PRELOADED BOLT ; CHS = CIRCULAR HOLLOW SECTION ;
EXC = EXECUTION CLASS ; MPI = MAGNETIC PARTICLE INSPECTION ; UT = ULTRASONIC TESTING ; MDD = MAXIMUM DRY DENSITY ;
GB = GROUND BEAM ; BB = BOND BEAM ; PC = STUMP COLUMN ; FFL = FINISHED FLOOR LEVEL.

STEELWORK STANDARD NOTES

1. STEEL FABRICATION AND ERECTION TO EN 1090-2 (EXC2). SHOP DRAWINGS TO BE APPROVED BEFORE FABRICATION.
2. ALL STEEL SHALL BE MARKED AND TRACEABLE (EN 10204 TYPE 3.1 CERTIFICATES FOR PRIMARY MEMBERS).
3. CUTTING: MACHINE-CUT OR PLASMA, EDGES GROUND SMOOTH; NO FLAME-CUT NOTCHES.
4. WELDED SECTIONS: WEB-TO-FLANGE WELDS BY SUBMERGED ARC, FULL-LENGTH. CAMBER = L/500 NOT REQUIRED. SWEEP <= L/1000.
5. HOLES: DRILLED, NOT FLAME-CUT. BOLT LENGTH SO THAT >= 2 THREADS PROJECT BEYOND THE NUT; WASHERS UNDER NUT AND HEAD FOR HV/HR.
6. ALL BOLTS IN END-PLATE JOINTS SHALL BE TIGHTENED IN A CROSS-PATTERN FROM THE CENTRE OUT; FINAL CHECK BY TORQUE WRENCH.
7. BASE PLATES SHALL BE GROUTED WITH NON-SHRINK GROUT AFTER FRAME ALIGNMENT AND TEMPORARY BRACING IS IN PLACE.
8. PURLIN SPLICES ARE LAPPED OVER RAFTERS (SUPPLIER STANDARD). SAG RODS Ø12 AT MID-SPAN WITH DIAGONAL TIES AT RIDGE.
9. ALL FIELD DRILLING / WELDING DAMAGE SHALL BE MADE GOOD WITH ZINC-RICH PAINT (2 x 40 um).
10. FIRE PROTECTION: NOT REQUIRED (SINGLE-STOREY, NO OCCUPANCY-BASED RATING ADVISED) - CONFIRM WITH LOCAL FIRE CODE.